Global Fuel Prices & Taxation

Project Report: Data Analytics with Tableau

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Project focus: global petrol, diesel and LPG prices, regional comparisons, income-level differences, taxation and price trends from 2020 to 2026.

Tableau Public Dashboard
Open Global Fuel Overview

Tableau Public Story
Open Global Fuel Prices Story

Prepared as a project report based on the Tableau workbook and screenshots supplied for this project.

1. Introduction

1.1 Project Overview

This project explores how fuel prices and taxation vary across countries, regions and income groups over the 2020–2026 period. Tableau is used to convert the dataset into a collection of interactive charts, a global petrol-price map, a dashboard and a guided story. The analysis covers petrol, diesel and LPG prices, Brent crude values, country-level tax percentages, region and income level.

The workbook contains views including:

- Global petrol price map for geographic comparison.
- Regional petrol, diesel and LPG price comparisons.
- Fuel price trends from 2020 to 2026.
- Fuel taxation comparison by country.
- Petrol price comparison by income level.
- Taxation versus petrol price scatter plot.

1.2 Purpose

The purpose is to make global fuel-price patterns easier to compare and interpret. The dashboard helps users identify regional differences, inspect taxation levels, compare income groups and observe how fuel prices change over time. It is intended as an exploratory data-analytics project rather than a real-time price service.

2. Ideation Phase

2.1 Problem Statement

Fuel-price data can be difficult to interpret when prices, tax percentages and country attributes are reviewed separately. Users need a consolidated view to compare regions and countries, see trends over time and explore whether price levels differ alongside taxation and income group.

Stakeholder	Role / perspective	Information need
Policy or energy analyst	Studies fuel affordability and public policy	Compare taxation and prices across countries and regions
Business / market analyst	Reviews operating costs and market differences	Identify high- and low-price regions and compare fuel types
General researcher or student	Explores global fuel data	Understand long-term trends and income-level differences

2.2 Brainstorming

The selected analysis themes were geographic distribution, regional fuel comparisons, price trends, country-level taxation, income-level comparisons and the relationship between tax percentage and petrol price. These themes are presented through individual worksheets and combined in the Global Fuel Overview dashboard and Global Fuel Prices Story.

3. Requirement Analysis

3.1 User Journey

Stage	User action	Dashboard support
Overview	Opens the dashboard for a broad picture	Global petrol price map and regional price charts
Comparison	Compares regions, countries and income levels	Bar charts for petrol, diesel, LPG and taxation
Analysis	Investigates time trends and possible relationships	Fuel-price trend line and tax-versus-petrol scatter plot
Exploration	Narrows the view using available filters	Region and income-level filters on relevant views
Sharing	Shares findings with others	Published Tableau Public dashboard and story links

3.2 Solution Requirements

Functional requirements

- Display country-level petrol prices on a global map.
- Compare average petrol, diesel and LPG prices across regions.
- Show fuel price trends across the 2020–2026 period.
- Compare tax percentages across countries.
- Compare petrol prices by income level and plot tax percentage against petrol price.
- Allow interactive exploration through available region, income-level and country filters.

Non-functional requirements

- Use clear chart titles, readable labels and consistent visual styling.
- Keep the dashboard suitable for interactive exploration.
- Publish the dashboard and story through Tableau Public for sharing.

3.3 Data Flow

Fuel-price dataset (Excel/CSV) → Tableau data source → field and aggregation setup → individual worksheets → Global Fuel Overview dashboard → Global Fuel Prices Story → Tableau Public.

3.4 Technology Stack

Layer	Tool / component
Dataset	Global fuel prices dataset containing country, date, region, income level, subsidy level, Brent crude and fuel-price/tax fields
Visualisation	Tableau Desktop
Interactive publication	Tableau Public
Documentation	Project report and dashboard/story screenshots

4. Project Design

4.1 Problem–Solution Fit

The project addresses the challenge of comparing multiple fuel types and taxation measures across a large number of countries. A dashboard combines the map, regional charts, trend view, income-level comparison and taxation analysis so users can move from a broad overview to more specific questions without reviewing separate raw tables.

4.2 Proposed Solution

- **Global Fuel Overview:** brings together the global petrol-price map, regional petrol comparison, petrol price by income level, fuel-price trends, taxation comparison, diesel and LPG regional comparisons, and taxation-versus-petrol scatter plot.
- **Interactive filters:** region and income-level filters support comparison on the relevant dashboard views.
- **Global Fuel Prices Story:** guides viewers through regional petrol prices, fuel-price trends, countries with high taxation, the relationship between taxation and petrol price, diesel and LPG comparisons, and petrol prices across income groups.
- **Publishing:** dashboard and story are shared using the supplied Tableau Public links.

4.3 Solution Architecture

Dataset → Tableau data source and measures → worksheets → dashboard composition → story points → Tableau Public. The project is based on descriptive visual analytics; no predictive model or live data-refresh mechanism is claimed in this report.

5. Project Planning & Scheduling

Phase	Activities	Output
1. Understand the problem	Define the questions around fuel prices, regions, income levels and taxes	Analysis scope
2. Prepare the data	Connect the dataset and check fields used by the worksheets	Tableau data source
3. Build worksheets	Create map, bar charts, trend chart and scatter plot	Individual views
4. Build dashboard	Arrange key views and filters into a combined overview	Global Fuel Overview
5. Build story and publish	Arrange story points and publish dashboard/story	Tableau Public links
6. Document	Capture screenshots and describe findings and limitations	Project report

6. Functional and Performance Testing

6.1 Validation Checklist

The following checks are based on the visible workbook and screenshots supplied. They describe visual and interaction checks; they are not a claim of automated load testing.

Test	Expected result	Evidence / status
Regional fuel-price charts	Regions appear with comparable average values	Visible in supplied worksheets; visual check
Taxation comparison	Country bars and tax values are readable	Visible in supplied worksheet; visual check
Income-level comparison	High, Middle and Low categories can be compared	Visible in supplied worksheet; visual check
Trend chart	Time series displays the 2020–2026 period	Visible in supplied worksheet; visual check
Dashboard filters	Available region and income-level filters can be used to narrow views	Controls visible in screenshot; interaction should be confirmed in Tableau Public
Public sharing	Dashboard and story open using the supplied links	Links included in this report; verify access in browser if needed

7. Results

7.1 Dashboard Output

The dashboard combines multiple views into one analytical page. The map supports geographic exploration; regional bar charts compare average petrol, diesel and LPG prices; the trend chart displays fuel-price movement over time; the taxation chart compares country-level percentages; the income-level chart compares average petrol prices; and the scatter plot places taxation percentage against petrol price.

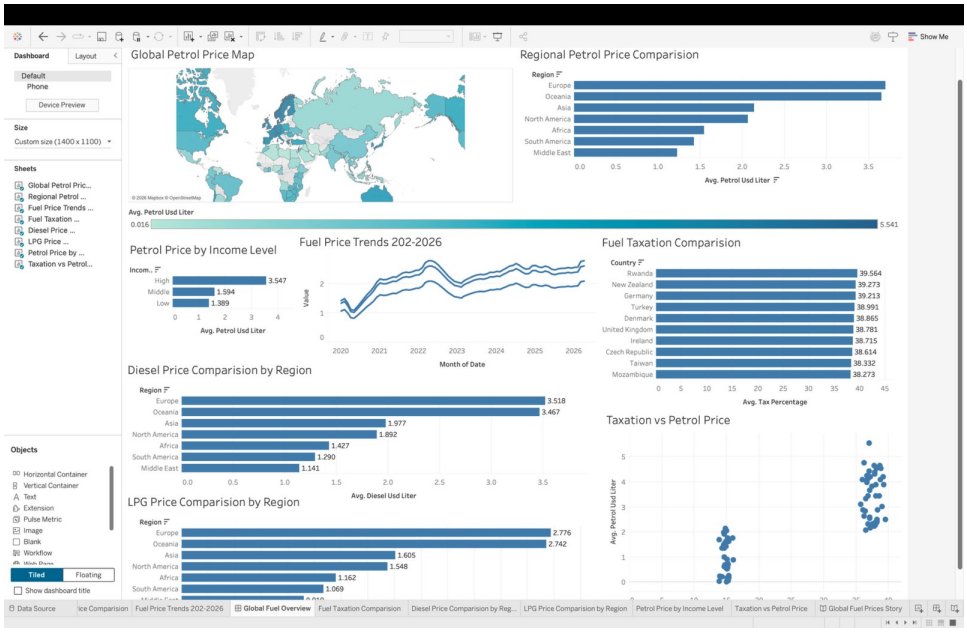


Figure 1. Global Fuel Overview dashboard screenshot supplied for the project.

7.2 Observations from the displayed views

- In the displayed regional comparisons, Europe and Oceania have the highest average values among the shown regions for petrol, diesel and LPG.

- The income-level chart shows a higher average petrol price for the High income group than for the Middle and Low groups in the displayed view.
- The taxation comparison shows high tax percentages for the displayed countries, including Rwanda, New Zealand and Germany.
- The scatter plot shows a visible cluster of countries around tax percentages in the mid-teens with lower petrol prices, and another cluster around the upper-thirties with a wider range of petrol prices. This is an observed pattern, not proof that taxation alone causes price differences.

7.3 Tableau Story Output

The story provides a guided sequence through the regional petrol comparison, fuel-price trends, countries with higher fuel taxation, the relationship between taxation and petrol price, regional diesel and LPG comparisons, and petrol prices across income groups.

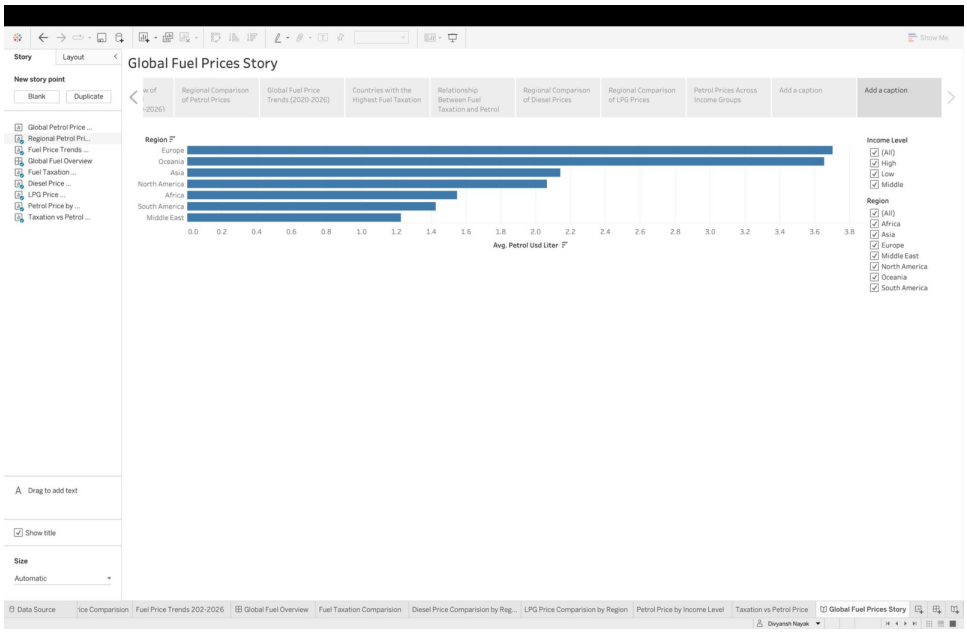


Figure 2. Global Fuel Prices Story screenshot supplied for the project.

8. Advantages & Limitations

Advantages

- Combines geographic, regional, temporal and taxation views in one project.
- Supports comparison across multiple fuel types and income groups.
- Uses interactive Tableau views and a shareable public dashboard/story.

Limitations

- The findings depend on the coverage, accuracy and definitions in the supplied dataset.
- The visualisations show descriptive patterns and do not establish causal relationships.
- Tableau Public content is publicly accessible; sensitive data should not be published there.
- The displayed values represent the dataset and selected aggregations, not necessarily current pump prices.

9. Conclusion

The Global Fuel Prices & Taxation project demonstrates how Tableau can organise country-level fuel data into a map, regional comparisons, time trends and taxation analysis. The combined dashboard and story make it easier to compare petrol, diesel and LPG prices, inspect income-level differences and explore how tax percentages vary with petrol prices. Conclusions should be interpreted in the context of the dataset and its aggregation choices.

10. Future Scope

- Add a documented data source and refresh date, with scheduled updates if a reliable source is available.
- Add country-level drill-downs and clearer units/currency definitions.
- Include inflation, exchange-rate, crude-oil and subsidy context for deeper interpretation.
- Add correlation statistics or regression only after checking data quality and assumptions.

11. Appendix

Tableau Public Dashboard: Global Fuel Overview

Tableau Public Story: Global Fuel Prices Story

Source dataset: Global fuel prices dataset used in the Tableau workbook. A public dataset URL was not provided in the supplied project details.

Workbook / code: Tableau workbook; no separate source-code repository was provided.